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1 – Context

Quantum processor scaling is currently limited by classical **room temperature electronics** :

- **Distance** → Increased latency
- **Wiring** → Excessive thermal load
- **Communication interface** → Limited parallelism

A promising solution is to move the readout and control electronics **on FPGA into the cryostat** ^[1]

2 – Challenges

Commercial FPGAs are not designed for this environment :

- × **Device models may be no longer valid**
- × **Embedded blocks may exhibit non-ideal behavior**
- × **External instrumentation access is limited**

How can we systematically validate FPGA designs in cryogenic environments, creating new models and guidelines for future works ?

3 – State of the Art

Essential FPGA blocks are tested → **Outperform their room-temperature ratings** ^[2]

More complex primitives remain untested → **Require high speed interfaces, exhaustive algorithms, or inaccessible instrumentations** ^[3]

Tested Blocks

- ✓ Flip-Flops
- ✓ Carry chains
- ✓ Multiplexers
- ✓ Look-Up Tables
- ✓ Inputs / Outputs
- ✓ Phase-locked loops

Untested Blocks

- × Serializers
- × Deserializers
- × Embedded memories
- × Hard processor cores
- × Digital signal processors

4 – Proposed approach

Adapt **Built-In Self-Test (BIST)** methodologies traditionally used for industrial silicon validation to an FPGA design flow

- ✓ At speed testing
- ✓ Exhaustive fault coverage
- ✓ Limited thermal load
- ✓ Limited external instrumentation

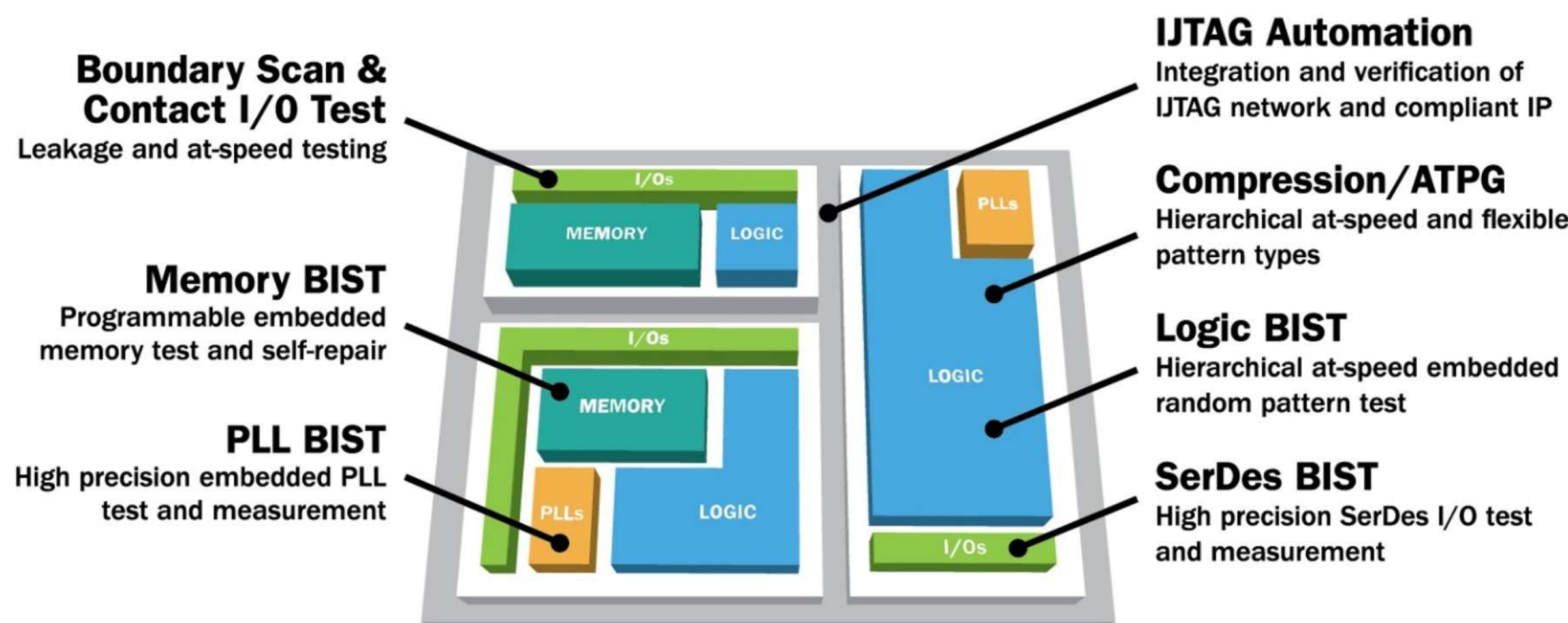


Fig. 1 Built-In Self Test Structures Available with Siemens EDA's Tessent ^[4]

5 – Methodology

Microchip's IGLOO2 FPGA ^[5]

- Flash technology → Non volatile + Low idle consumption
- Phased-Locked-Loop (PLL) → Up to 400 MHz
- 1 Kbits μ SRAM blocks → Up to 400 MHz

Tessent's Memory BIST

- Robust fault coverage
- JTAG-USB Adaptor
- 5 wires at 3 MHz

Arbitrary Wave Generator ^[6]

- PLL reference input
- 10 to 50 MHz

Ice Oxford Dry-Ice Cryostat ^[7]

- Control from 300 down to 10 K

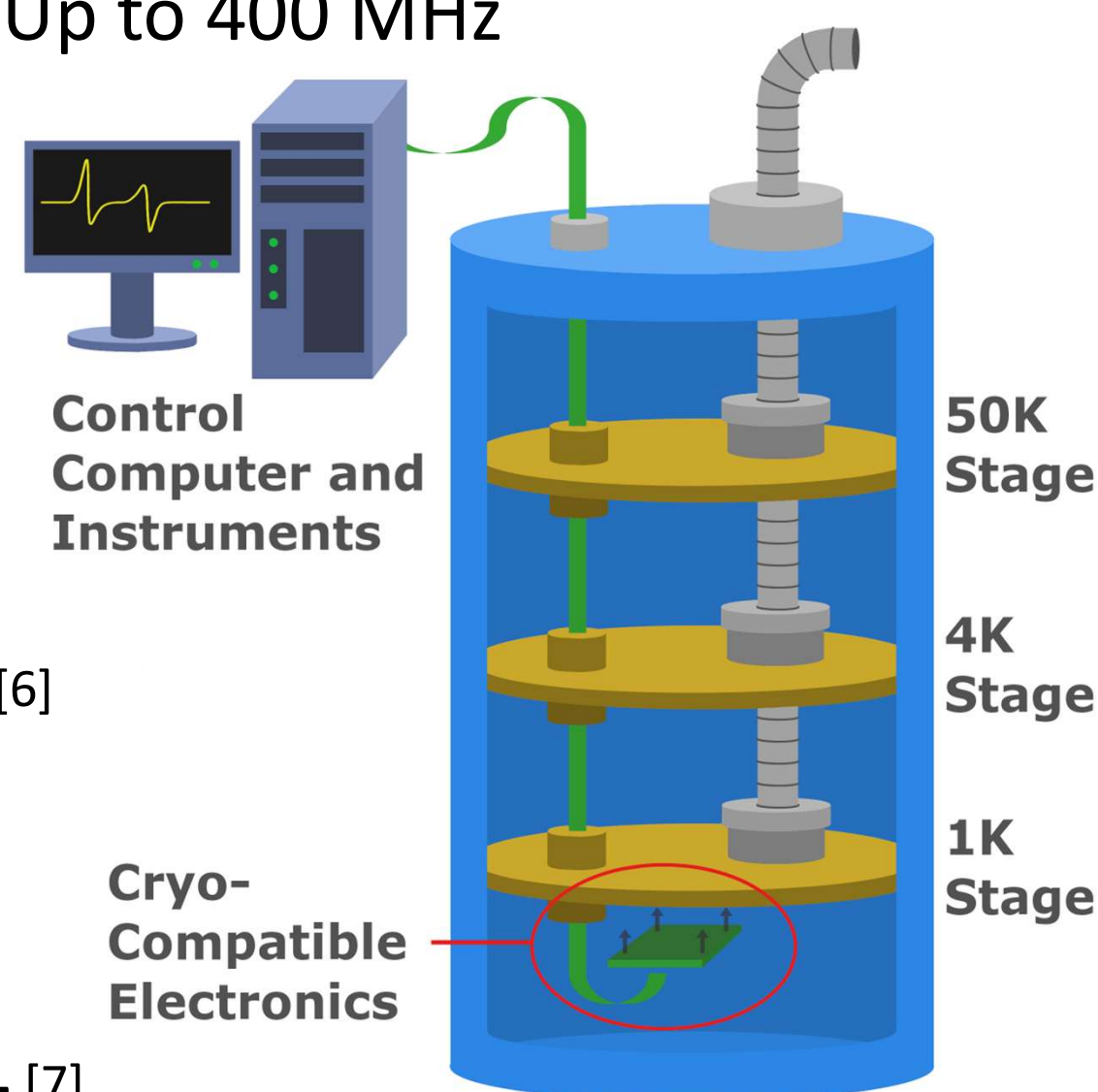


Fig. 2 Experimental Setup

6 – Results

The μ SRAM is functional within specs down to 10 K
Memory Built-In Self-Test Results

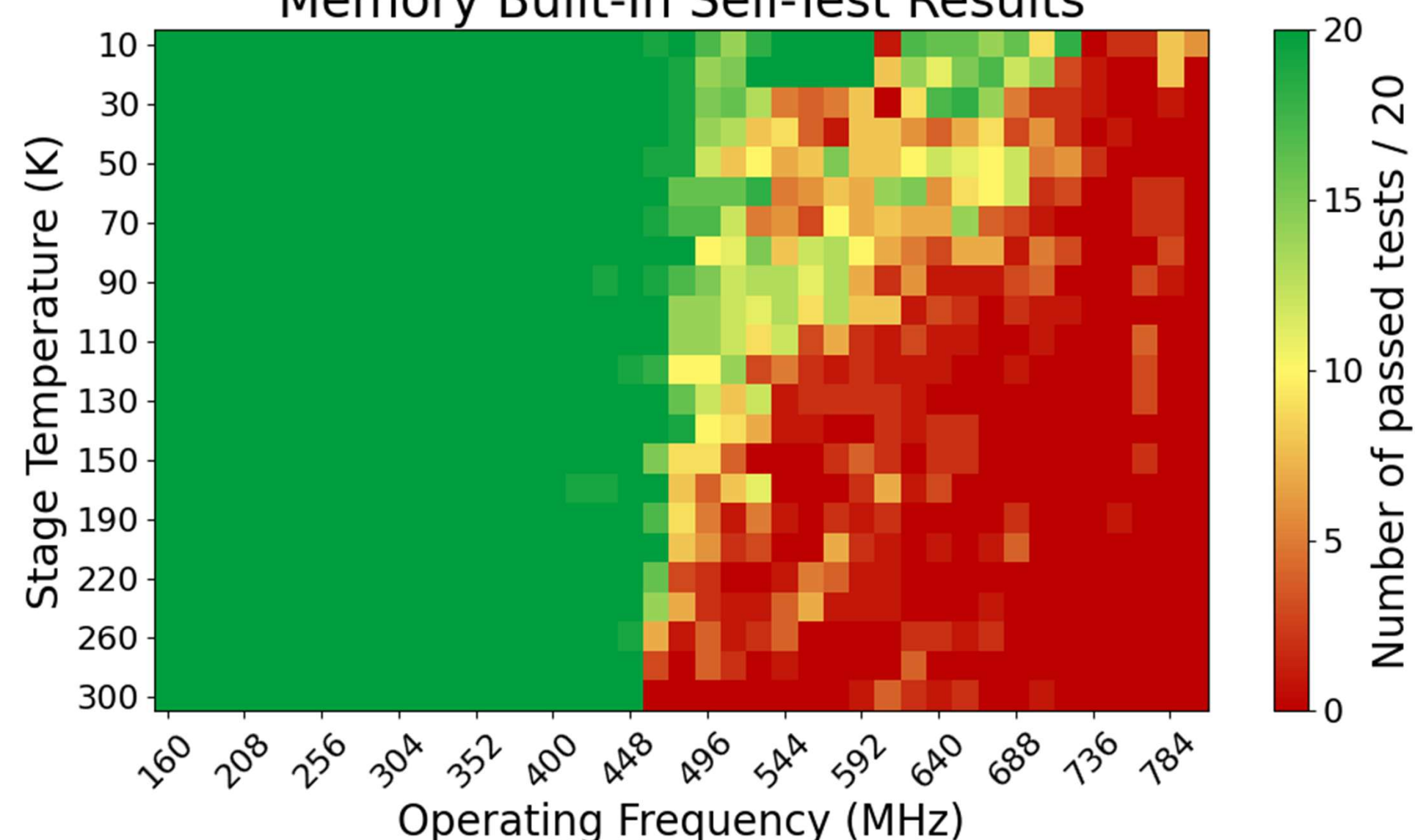


Fig. 3 Memory BIST Heat Map on an IGLOO2's uRAM

When reaching less than 2 K →

The μ SRAM passed 130 out of 151 tests at 736 MHz

7 – Conclusion

What could be improved ?

- PLL operation within specs
- Two wire JTAG protocol

Future Works ?

- Characterize more primitives
- Integrate "in field" testing

Develop, test, characterize, and calibrate a fully integrated FPGA-based qubit control and readout systems

8 – Acknowledgements



Scan me for sources !

